CODEFORCES (#96A)

Football: Detecting Dangerous Player Formations

Given a string of 0s and 1s representing players from two teams, return YES if at least seven consecutive players belong to the same team; otherwise return NO.

Input constraints:

- $1 \leq |s| \leq 100$ - Each character is 0 or 1 **- Both teams must be represented**

Expected behavior:

Return YES exactly when 0000000 or 1111111 appears in the string.

Natural-Language Contracts (NLCs)

Preconditions

Pre 1. The input string is non-empty.
Pre 2. The input contains only 0 and 1.
Pre 3. The input length does not exceed 100.
***Pre 4.* Both 0 and 1 must occur in the input.**

Postconditions

Post 1. Return YES if seven consecutive identical characters occur.
Post 2. Otherwise, return NO.

Contract Integration (Contract Explicit)

```
import icontract
@icontract.require(lambda s: len(s) > 0)
@icontract.require(lambda s: all(c in "01" for c in s))
@icontract.require(lambda s: len(s) ⩽ 100)
@icontract.require(lambda s: "0" in s and "1" in s)
@icontract.ensure(
    lambda s, result: result in ("YES", "NO")
)
@icontract.ensure(
    lambda s, result:
        (result == "YES") ==
        ("0000000" in s or "1111111" in s)
)
```

Executable Contracts (ECs)

```
def solve(s):
    if "0000000" in s or "1111111" in s:
        return "YES"
    return "NO"
```

NLC Guided Code (Contract Implicit)

Contract Checker Inputs

Valid Input: `1000000001`
`Expected: YES`

Invalid Input: `0000000`
`Expected: Pre-4 Violation`

Contract Implementation

Implicit	Explicit
✓ YES	✓ YES
! YES	✓ **ViolationError**